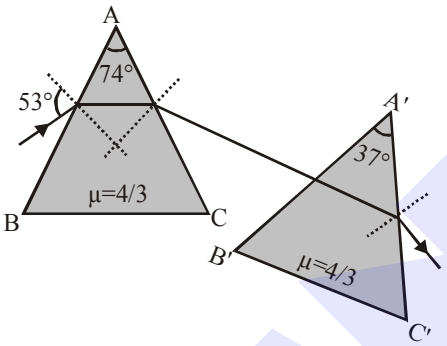
SECTION-I

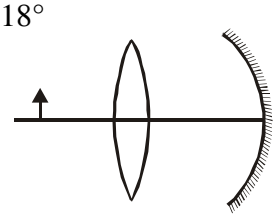
Single Correct Answer Type **4 Q. [3 M (–1)]**

1. A light ray is incident on face AB of prism ABC as shown in figure. The second prism is kept in such a manner that emergent ray from prism ABC is falling normally on face $A'B'$ of prism $A'B'C'$. The net deviation produced by the optical system of the two prisms is :-



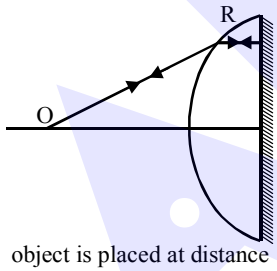
(A) 48° (B) 112° (C) 32° (D) 18°

2. Figure shows a thin converging lens for which the focal length is 5 cm. The lens is in front of a concave spherical mirror of radius R = 30 cm. If the lens and mirror are 20 cm apart and an object is placed 15 cm to the left of the lens, determine the approximate distance of the final image from the lens (in cm)

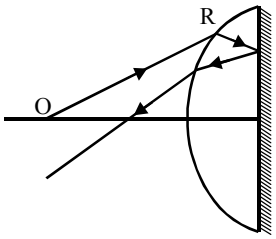


(A) 5.3 cm (B) 4.6 cm
(C) 6.1 cm (D) 12.7 cm

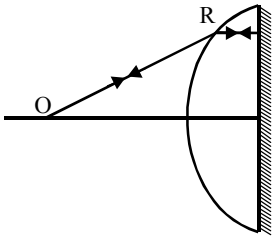
3. A thin plano-convex glass lens (μ = 1.5) has its plane surface reflecting and R is the radius of curvature of curved part, then which of the following ray diagram is true for an object placed at O ?

(A)


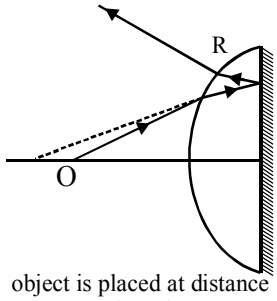
object is placed at distance 2R from lens

(B)


object is placed at distance 2R from lens

(C)


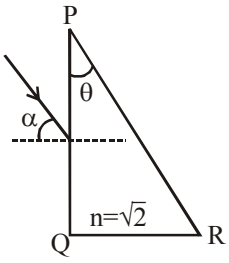
object is placed at distance 3R from lens

(D)


object is placed at distance 2R from lens

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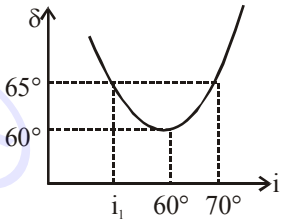
4. A parallel beam of light is incident from air at an angle α on the side PQ of a right angled triangular prism of refractive index $n = \sqrt{2}$. Light undergoes total internal reflection in the prism at the face PR when α has a minimum value of 45°. The angle θ of the prism is :



(A) 15°
(B) 22.5°
(C) 30°
(D) 45°

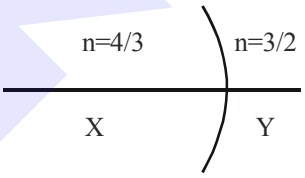
Multiple Correct Answer Type 11 Q. [4 M (–1)]

5. The angle of deviation (δ) vs angle of incidence (i) is plotted for a prism. Choose the **CORRECT** statement(s).



(A) The angle of prism is 60°
(B) The refractive index of the prism is n = √3
(C) For deviation to be 65° the angle of incidence $i_1 = 55°$
(D) for minimum deviation, the angle of emergence e = 60°.

6. A concave spherical surface of radius of curvature 10 cm separates two mediums X and Y of refractive indices 4/3 and 3/2 respectively. Centre of curvature of the surface lies in the medium X. An object is placed in medium X. Choose **INCORRECT** option(s) :-



(A) Image is always real
(B) Image is real if the object distance is greater than 90 cm.
(C) Image is always virtual
(D) Image is virtual only if the object distance is less than 90 cm.

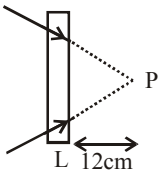
7. The principal axis of an optical device is along y = –1. If the image of a small body placed at (–30,3) is formed at a point (60, –3), then the optical device may be :-

(A) A convex lens of focal length 20 cm
(B) A concave mirror of focal length 60 cm
(C) A concave lens of focal length 20 cm
(D) A convex mirror of focal length 60 cm

8. In displacement method, the distance between object and the screen is 96 cm. The ratio of length of two images formed by a converging lens placed between them is 4. Then

(A) ratio of the length of object to the length of shorter image is 2.
(B) distance between the two positions of the lens is 32 cm.
(C) focal length of the lens is 64/3 cm.
(D) when the shorter image is formed on screen, distance of the lens from the screen is 32 cm.

9. A beam of light converges to a point P. A device L is placed in the path of the convergent beam 12 cm from P as shown. Choose the correct option(s) :-



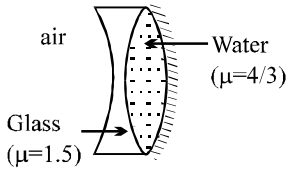
(A) If L is a convex lens of focal length 20 cm, the beam converges at a point at 4.5 cm distance from P
(B) If L is a convex lens of focal length 20 cm, the beam converges at a point at 10 cm distance from lens
(C) If L is a concave lens of focal length 16 cm, the beam converges at a point at 48 cm distance from lens
(D) If L is a plane mirror the beam converges at a point at 24 cm distance from P.

10. Variation of magnification (m) produced by a thin converging lens versus distance (v) of image from pole of the lens is plotted. Which of the following statements is/are correct?

(A) Focal length of the lens is equal to intercept on v-axis.
(B) Focal length of the lens is equal to inverse of the magnitude slope of the line.
(C) Magnitude of intercept on m-axis is equal to unity.
(D) Magnitude of intercept on v-axis is equal to unity.

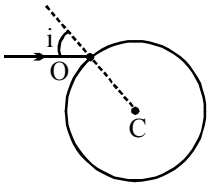
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11. The radius of curvature of the left and right surface of the concave lens are 10cm and 15cm respectively. The radius of curvature of the mirror is 15cm.



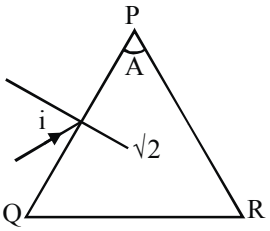
(A) equivalent focal length of the combination is –18cm.
(B) equivalent focal length of the combination is +36cm.
(C) the system behaves like a concave mirror.
(D) the system behaves like a convex mirror.

12. A spherical transparent medium of refractive index n is placed in air. A thin parallel beam of light of monochromatic wavelength λ is incident at a point *O* of the sphere at an angle of incidence *i*. Then which of the following statements will be **true** for this optic system?



(A) For the ray to come out of the sphere after refraction through it, the incident angle i should be less than the critical angle C for the medium of the sphere.
(B) The angle of emergence θ (made with the normal) for the ray will be equal to the incident angle *i* for all values of *i*.
(C) If angle $i \simeq 90°$ (grazing incidence) the emergent ray will come out grazing the surface.
(D) If the angle of incidence i is such that the angle of refraction at *O* is equal to the critical angle *C* for the medium of sphere, the emergent ray will be parallel to the incident ray.

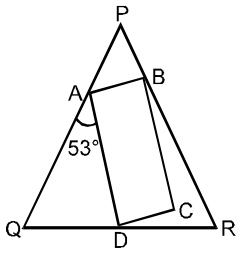
13. A prism of refractive index $\sqrt{2}$ and apex angle A is shown. Light is incident from *PQ* side at angle of incidence *i* $(0 < i \leq 90°)$



(A) if $A = 40°$ then light incident at all angles will emerge from surface *PR*.
(B) if $A = 80°$ then light incident at some angles will emerge from surface PR and at some other angles light will suffer TIR at surface PR.
(C) if $A = 100°$ then light incident at all angles will be reflected back from surface *PR*.
(D) whatever is the value of *A*, light will emerge from the surface *PR* for some value of *i*.

14. Optical axis of a thin equi-convex lens is the X-axis. The co-ordinate of a point object and its image are (–20 cm, 1 cm) and (25 cm, –2 cm) respectively
(A) the lens is located at x = 5 cm
(B) the lens is located at x = –5 cm
(C) the focal length of the lens is 10 cm
(D) the focal length of the lens is 15 cm

15. A rectangular cavity ABCD is carved inside an equilateral prism PQR of refractive index $\sqrt{2}$ shown in the figure. For a ray entering at face PQ and emerging at face PR, without passing through AB and CD:



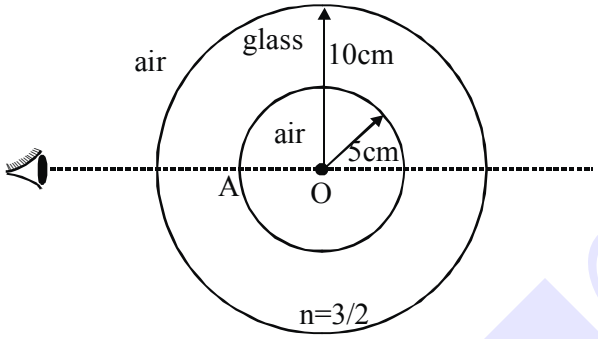
(A) there will not be any effect of cavity on δ
(B) there may be effect of cavity on δ
(C) for cavity filled with water $\delta_{min} = 30°$
(D) for cavity filled with air $\delta_{min} = 30°$

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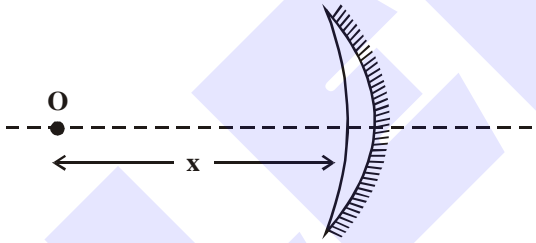
SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **4 Q. [4 M (0)]**

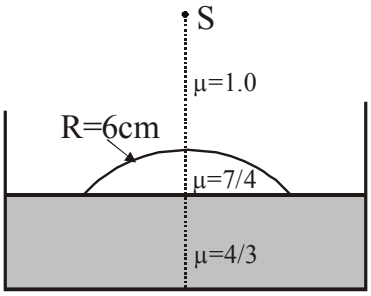
16. A hollow glass sphere of inner radius 5 cm & outer radius 10 cm is viewed as shown in figure. If refractive index of glass is 3/2, find distance (in cm) between *A* & *O* as seen by the observer?



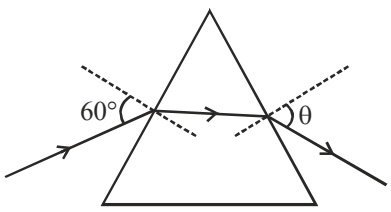
17. Radii of curvature of a concavo-convex lens (refractive index = 1.5) are 40 cm and 20 cm as shown. The convex side is silvered. The distance x (in cm) on the principal axis where an object is placed so that its image is created on the object itself, is given as 4β. Find the value of β.



18. Water (with refractive index = $\frac{4}{3}$) in a tank is 18 cm deep. Oil of refractive index $\frac{7}{4}$ lies on water making a convex surface of radius of curvature 'R = 6 cm' as shown. Consider oil to act as a thin lens. An object 'S' is placed 24 cm above water surface. The location of its image is at 'x' cm above the bottom of the tank. Then 'x' is



19. A monochromatic beam of light is incident at 60° on one face of an equilateral prism of refractive index n and emerges from the opposite face making an angle θ(n) with the normal (see the figure). For n = $\sqrt{3}$ the value of θ is 60° and $\frac{d\theta}{dn} = m$. The value of m is

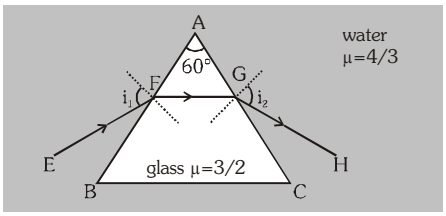


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SECTION-IV

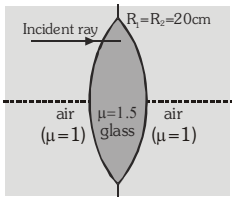
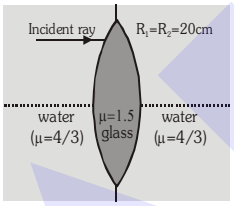
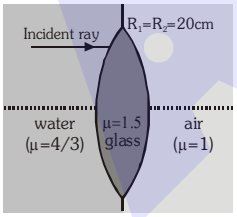
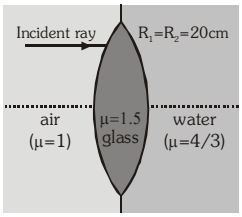
Matrix Match Type (4 × 5) **3 Q. [8 M (for each entry +2(0))]**

20. Consider an equilateral prism ABC of glass $\left(\mu = \frac{3}{2}\right)$ placed in water $\left(\mu = \frac{4}{3}\right)$



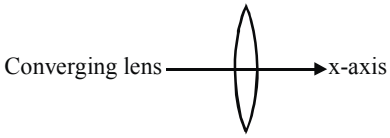
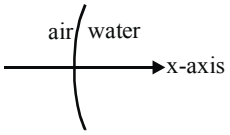
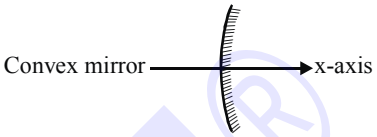
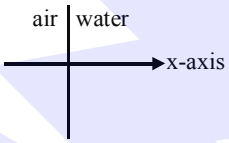
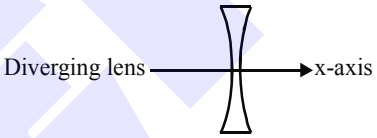
	Column–I		**Column–II**
(A)	FG is parallel to BC	(P)	maximum deviation
(B)	$i_1 = 90°$	(Q)	minimum deviation
(C)	$i_1 = i_2 = \sin^{-1}\left(\frac{9}{16}\right)$	(R)	TIR will take place at surface AC
(D)	EF is perpendicular to AB	(S)	No TIR will take place at surface BC
		(T)	Can't say anything

21.

	Column I (optical system)		**Column II (focal length)**
(A)		(P)	80 cm
(B)		(Q)	40 cm
(C)		(R)	30 cm
(D)		(S)	20 cm

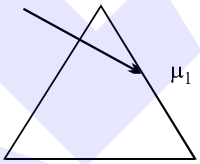
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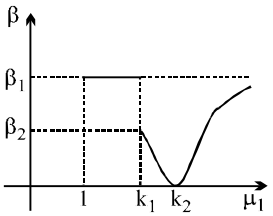
22. In column I certain combination of small object and image are given. Match them with possible refracting / reflecting body in column II. In all cases, object is perpendicular to x-axis. Object can't be at zero distance from the interface.

Column I		**Column II**
(A) Virtual image of virtual object	(P)	
(B) Real image of real object	(Q)	Light is incident from air on the spherical surface 
(C) Erect real image of same size as object	(R)	
(D) Erect and virtual image of larger size	(S)	Light is incident from right to left. 
	(T)	

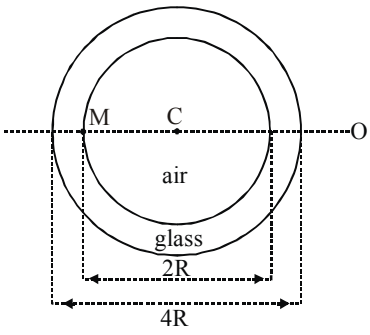
Subjective Type 6 Q. [4 M (0)]

23. The diagram shows an equilateral prism. The medium on one side of the prism is μ_1. The refractive index of the prism is $\mu = 4/\sqrt{3}$. The diagram shows the variation of the magnitude of angle of deviation with respect to μ_1? Consider the light ray to be normally incident on the first face. Find the value of β_1, β_2, k_1 & k_2.



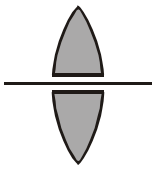


24. A hollow sphere of glass of R.I. n has a small mark M on its interior surface which is observed by an observer O from a point outside the sphere. C is the centre of the sphere. The inner cavity (air) is concentric with the external surface and thickness of the glass is every where equal to the radius of the inner surface. Find the distance by which the mark will appear nearer than it really is in terms of n and R assuming paraxial rays.

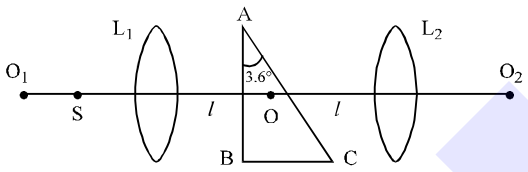


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25. A point object is placed at a distance of 0.3m from a convex lens (focal length 0.2 m) cut into two halves each of which is displaced by 0.0005 m as shown in the figure. Find the position of the image. If more than one image is formed, find their number and the distance between them.

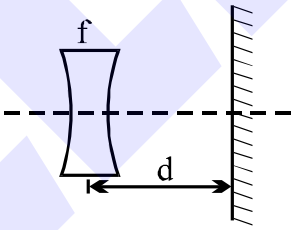


26. A thin right angled glass prism of cross-section ABC with a very small refracting angle $\alpha = 3.6°$ is placed between the convex lenses along a common axis O_1O_2 (as shown in figure)

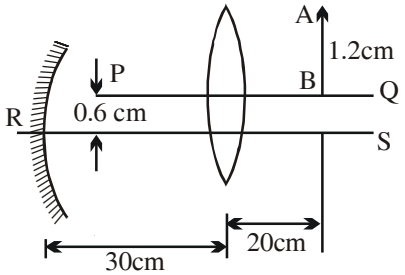


The lenses are placed at equal distances l = 30 cm from the prism. The refractive index of the prism glass is $\mu = 1.5$. The surface AB does not reflect any light energy while the surface AC is partially reflecting and partially refracting. A point source of light S is placed at a distance of 20 cm to the left of the lens L_1 as shown. Considering the rays that are closed to the axis O_1O_2, find the possible position of the images(s) formed by this system. Show the image on a proper ray diagram. The focal lengths of the lenses L_1 and L_2 are 20 cm each. The face AB of the prism is normal to the axis O_1O_2.

27. A diverging lens of focal length 10 cm is placed 10 cm in front of a plane mirror as shown in the figure. Light from a very far away source falls on the lens. Find the image of source due to plane mirror (before hitting lens again) at a distance from mirror?



28. A convex lens of focal length 15 cm and a concave mirror of focal length 30 cm are kept with their optic axes *PQ* and *RS* parallel but separated in vertical direction by 0.6 cm as shown. The distance between the lens and mirror is 30 cm. An upright object *AB* of height 1.2 cm is placed on the optic axis *PQ* of the lens at a distance of 20 cm from the lens . If *A′B′* is the image after refraction from the lens and reflection from the mirror, find the distance of *A′B′* from the pole of the mirror and obtain its magnification. Also locate positions of *A′* and *B′* with respect to the optic axis *RS*. **[IIT-JEE 2000]**



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GR # GEOMETRICAL OPTICS-II **ANSWER KEY**

SECTION-I

Single Correct Answer Type **4 Q. [3 M (–1)]**

1. Ans. (A) **2. Ans. (A)** **3. Ans. (A)** **4. Ans. (A)**

Multiple Correct Answer Type **11 Q. [4 M (–1)]**

5. Ans. (A,B,C,D) **6. Ans. (A,B,D)** **7. Ans. (A,B)** **8. Ans. (A, B, C, D)** **9. Ans. (C, D)**
10. Ans. (A,B,C) **11. Ans. (A,C)** **12. Ans. (B,C)** **13. Ans. (A,B,C)** **14. Ans. (B,C)**
15. Ans. (A,C,D)

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **4 Q. [4 M (0)]**

16. Ans. 6 **17. Ans. 4** **18. Ans. 2** **19. Ans. 2**

SECTION-IV

Matrix Match Type (4 × 5) **3 Q. [8 M (for each entry +2(0)]**

20. Ans. (A) → (Q,S); (B) → (P,S); (C) → (Q,S); (D) → (S)
21. Ans. (A) → (S) ; (B) → (P) ; (C) → (R) ; (D) → (Q)
22. Ans. (A) → (P,R,S,T) ; (B) → (P,Q) ; (C) → (S) ; (D) → (P,Q)

Subjective Type **6 Q. [4 M (0)]**

23. Ans. $60°, 30°, 2, \frac{4}{\sqrt{3}}$ **24. Ans. (n - 1)R/(3n - 1)**

25. Ans. [Number of images = 2, distance between them = 0.003 m]

26. Ans. $y_1 = f\,\delta_1 = 20 \times \frac{\pi}{100}$ cm = π/5 cm ; $y_2 = f\,\delta_2 = 20 \times \frac{3\pi}{50}$ cm = 1.2π cm

27. Ans. 20 cm behind the mirror

28. Ans. 15 cm right of the mirror, magnification = 1.5; (–15 cm, –1.5 cm; – 15 cm, 0.3 cm)

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